Analysis of gene regulatory properties underlying trait pleiotropy

Aitor González*

TAGC Laboratory, Aix-Marseille Univ, INSERM UMR1090, 13288 Marseille, France

*Email: aitor.gonzalez@univ-amu.fr

Background and Objectives

Background

1. Genome-wide association studies (GWAS) map genotypes to traits.
2. Here we use this GWAS database: The IEU OpenGWAS project `https://gwas.mrcieu.ac.uk`
3. The large number of available GWAS have allowed to define pleiotropic regions in the genome [2].
4. Most GWAS loci are non-coding but the gene regulatory basis of pleiotropy is not well understood.
5. Expression quantitative trait loci (eQTL) map genotypes to expression phenotypes.
6. Here we use this eQTL database: The EBI eQTL Catalogue `https://www.ebi.ac.uk/eqtl`
7. Databases of GWAS and eQTL studies allow large colocalization analysis [1].
8. The R package coloc performs colocalization tests between genetic traits: `https://CRAN.R-project.org/package=coloc`

Objectives

1. To develop an eQTL/GWAS variant colocalization pipeline
2. To investigate gene regulatory properties underlying trait pleiotropy

Results

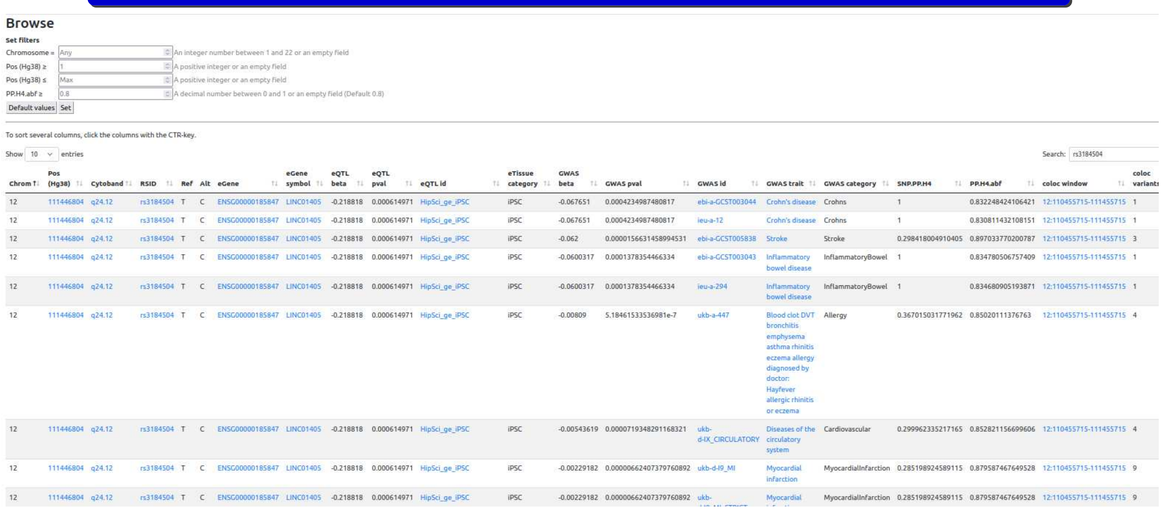

Figure 1: Web interface with the results of the colocalization pipeline for the most pleiotropic variant rs3184504.

Chrom.	Cytoband	Pos (hg38)	Variant	GWAS Categories
12	q24.12	111,446,804	rs3184504	AllergicRhinitis, Allergy, CancerGenitalFemale, Cardiovascular, CoronaryArtery, Crohns, Heart, InflammatoryBowel, MyocardialInfarction, Stroke, StrokeIschaemic
12	q24.12	111,569,952	rs653178	AllergicRhinitis, Allergy, CancerGenitalFemale, Cardiovascular, CoronaryArtery, Heart, InflammatoryBowel, MyocardialInfarction, Stroke, StrokeIschaemic
12	q24.12	111,427,245	rs7310615	AllergicRhinitis, Allergy, CancerGenitalFemale, Cardiovascular, CoronaryArtery, Heart, MyocardialInfarction
12	q24.12	111,535,554	rs597808	AllergicRhinitis, Allergy, CancerGenitalFemale, Cardiovascular, CoronaryArtery, Heart, MyocardialInfarction
12	q24.12	111,621,753	rs11065979	Aging, AllergicRhinitis, Allergy, Asthma, Stroke, StrokeIschaemic
15	q26.1	90,884,967	rs2071382	Angina, Cardiovascular, CoronaryArtery, Heart, Hypertension, Schizophrenia
1	p13.3	109,274,968	rs12740374	Angina, Cardiovascular, CoronaryArtery, Heart, MyocardialInfarction
1	p13.3	109,275,684	rs629301	Angina, Cardiovascular, CoronaryArtery, Heart, MyocardialInfarction
3	q23	141,406,344	rs1344674	Allergy, Asthma, CancerBreast, CancerProstate, Height
5	q31.1	132,485,305	rs17622656	Allergy, Asthma, Cardiovascular, Hypertension, UlcerativeColitis
7	p21.1	19,009,765	rs2107595	Angina, Cardiovascular, Heart, Stroke, StrokeIschaemic
11	q13.5	76,588,605	rs11236797	AllergicRhinitis, Asthma, Crohns, InflammatoryBowel, UlcerativeColitis
15	q26.1	90,879,313	rs8039305	Angina, Cardiovascular, CoronaryArtery, Heart, Hypertension
15	q26.1	90,880,313	rs6224	Angina, Cardiovascular, Heart, Hypertension, Schizophrenia

Table 1: Pleiotropic regions involving 5 or more GWAS categories. Genomic coordinates are given for the hg38 assembly.

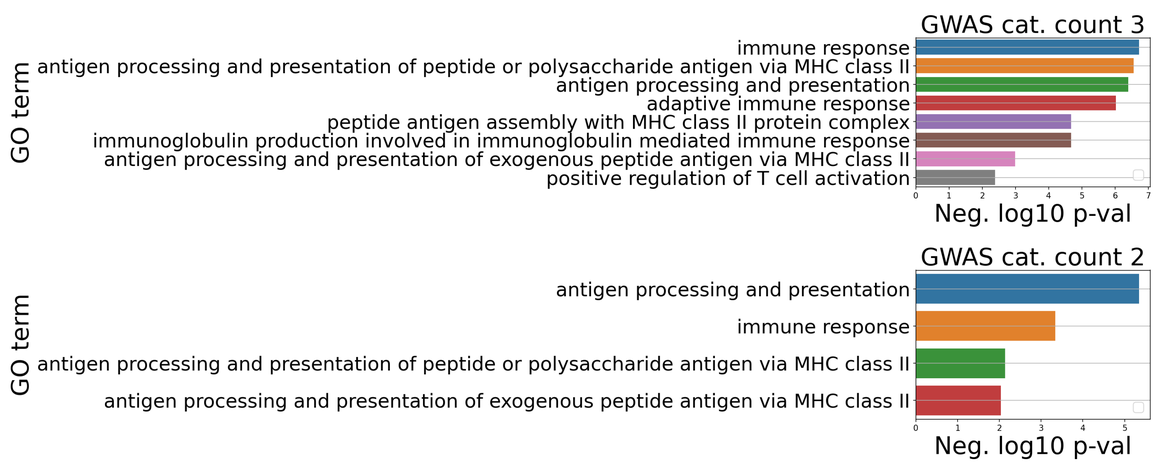

Figure 2: Gene ontology analysis of genes associated with eQTLs (eGenes) of variants associated to three (**Upper**) and two (**Bottom**) GWAS categories compared to eGenes of variants associated to one GWAS category.

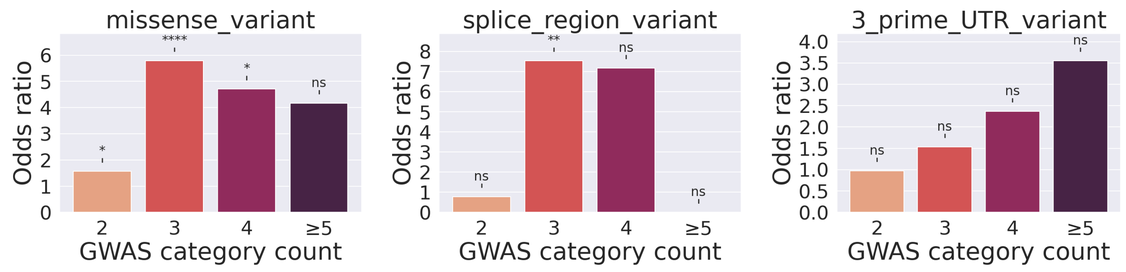

Figure 3: Variant effect predictor (VEP) analysis as a function of the number of GWAS catategory number for three different consequences: Missense variant (**Left**), splice region variant (**Center**) and 3' UTR variant (**Right**)

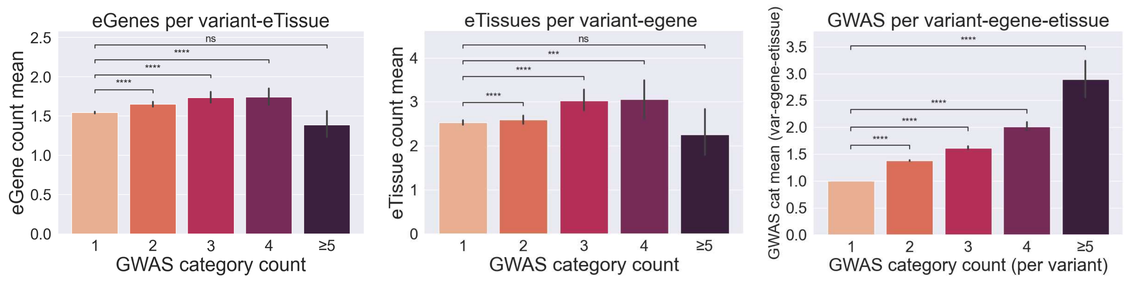

Figure 4: Count of eGenes per variant-tissue (**Left**), samples of eQTLs (eTissues) per variant-eGene (**Center**) and GWAS categories per variant-eGene-eTissue (**Right**).

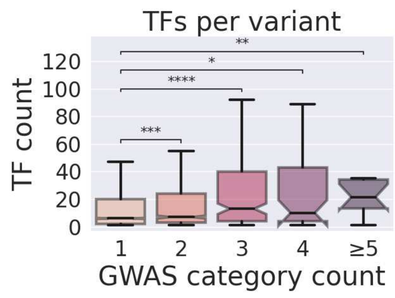

Figure 5: Binding count of transcription factors in the region surrounding pleiotropic variants (Radius 50 bp) as a function of the GWAS category count.

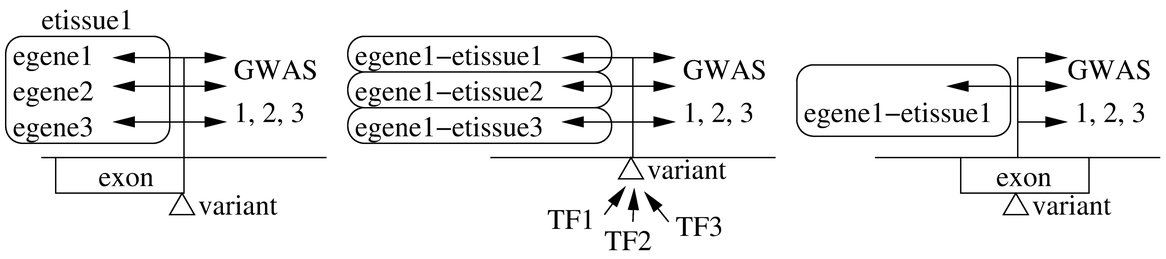

Figure 6: Model of regulatory variant pleiotropy. I have investigated three possible mechanisms of pleiotropy. (**Left**), Pleiotropic variants have more eGenes that result in more functions and more phenotypes. This might arise from an enrichment of pleiotropic variants in splicing or 3' UTR regions. (**Center**), I have also found that eGenes of pleiotropic variants are active in more etissues which result in more GWAS phenotypes. This might be explained from variants being bound by more transcription factors. (**Right**), Triplets of variant-eGene-eTissues are associated with more GWAS pheno- types, which directly affect the number of GWAS phenotypes. I have found that this might be explained by an enrichment of missense alleles.

Conclusions

- We have developed a pipeline of eQTL/GWAS colocalization based on 127 eQTL studies and 150 GWAS (**Fig. 1**).
- These colocalization results are available in an easy to use web site (**Fig. 1**).
- We have computed a list of pleiotropic eQTLs (**Table 1**).
- eGenes of pleiotropic variants are enriched in immune related functions (**Fig. 2**).
- Pleiotropic variants are enriched in missense and splice region VEP effect consequences (**Fig. 3**).
- Pleiotropic eQTLs have more eGenes per variant-eTissue, are active in more eTissues per variant-eGene and are associated to more GWAS categories per variant-eGene-eTissue (**Fig. 4**).
- Pleiotropic variants bind more transcription factors (**Fig. 5**).
- We propose a model explaining the hypotheses underlying the pleiotropy for eQTLs (**Fig. 6**).

References

[1] Zepeng Mu, Wei Wei, Benjamin Fair, Jinlin Miao, Ping Zhu, and Yang I. Li. The impact of cell type and context-dependent regulatory variants on human immune traits. *Genome Biology*, 22(1), apr 2021.

[2] Kyoko Watanabe, Sven Stringer, Oleksandr Frei, Maša Umićević Mirkov, Christiaan de Leeuw, Tinca J. C. Polderman, Sophie van der Sluis, Ole A. Andreassen, Benjamin M. Neale, and Danielle Posthuma. A global overview of pleiotropy and genetic architecture in complex traits. *Nature Genetics*, 51(9):1339–1348, aug 2019.